THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A mushroom growing on a log gets its food from: 1

(a) dead wood **(b)** sunlight **(c)** air **(d)** insects

2. In a pinhole camera, the size of image increases when the screen is moved: 1

(a) upside down **(b)** sideways

(c) towards the hole **(d)** away from the hole

3. In a partial eclipse, the observer is in the: 1

(a) total darkness **(b)** full sunlight **(c)** umbra **(d)** penumbra

4. A leaf covered partly with black paper is tested with iodine. The covered part: 1

(a) turns red **(b)** remains brown **(c)** turns blue-black **(d)** turns green

5. A ray falls along the normal. The angle of reflection is: 1

(a) 180° **(b)** 45° **(c)** 90° **(d)** 0°

6. In June, which hemisphere has long days? 1

(a) northern **(b)** southern **(c)** both equally **(d)** neither

7. The root hairs increase the surface for absorption of: 1

(a) light **(b)** oxygen only

(c) water and minerals **(d)** food

8. The shadow of a pole is longest at: 1

(a) 11 a.m. **(b)** midday

(c) sunrise or sunset **(d)** noon

9. A new moon rises at about: 1

(a) midnight **(b)** sunset **(c)** sunrise **(d)** 3 p.m.

10. A pitcher plant in nitrogen-poor soil survives by: 1

(a) absorbing more water **(b)** growing taller

(c) sleeping **(d)** trapping insects

11. A bigger hole in a pinhole camera makes the image: 1

(a) smaller **(b)** inverted

(c) sharper and dimmer **(d)** blurred and brighter

12. A lunar eclipse lasts longer than a solar eclipse because: 1

(a) the Sun is small **(b)** the Moon is large
(c) the Earth's shadow is large **(d)** the Moon moves fast

13. If carbon dioxide is absorbed by potassium hydroxide in a closed jar and a leaf inside is tested: 1

(a) nothing **(b)** more starch
(c) green colour **(d)** no starch is formed

14. A girl stands 1.5 m from a mirror. She walks 0.5 m towards it. The distance between her and her image now is: 1

(a) 2 m **(b)** 3 m **(c)** 1 m **(d)** 1.5 m

15. In December, the southern hemisphere has: 1

(a) summer **(b)** spring only
(c) the same as north **(d)** winter

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are the Moon's craters preserved for a long time? 2

17. Why are leguminous plants good for crop rotation? 2

18. How do mirrors in a periscope make the image erect? 2

19. Explain why the temperature is lower in winter. 2

20. Explain why a plant kept in the dark for a few days loses starch from the leaves. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why we see a smooth and bright image in a plane mirror but not in a wall. 3

22. Why do astronauts jump higher on the Moon? 3

23. Explain how the transport of water and food in plants differ. 3

24. A girl 1.4 m tall stands 3 m from a mirror. State the height of her image and the distance of the image from her. 3

OR

Explain how the Moon's phases can be shown with a ball and a lamp. 3

25. Explain why the photosynthesis rate is high at noon and low at dawn. 3

OR

Explain why the surface of a calm lake acts as a mirror and a rough lake does not. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain how a solar and a lunar eclipse occur using the ideas of shadow. 5

OR

Explain why the Earth is slightly flattened at the poles (qualitative) and the effect of rotation. 5

27. Describe the experiment to show that light is necessary for photosynthesis, with the steps, observation and conclusion. 5

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OR

Explain how the size of the image in a pinhole camera changes with distance of the object and the distance between hole and screen. 5

END OF QUESTION PAPER **Total: 50 Marks**

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